# Interest Rate Models

## Advanced Financial Modeling

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UNIVERSITÀ DI SIENA 1240

1 Introduction

2 The HJM Framework

3 Extended Short Rate Models

4 Gaussian Dynamics

## Suggested References

Everything will be covered during the course and provided to you as slides, notes or Python notebooks.
If you want more, check the following:

- D. Brigo, F. Mercurio (2006) Interest Rate Models - Theory and Practice (2nd edition)
- L. Andersen, V. Piterbarg (2010) Interest Rate Models. Vol.1, Vol.2, Vol.3.
- T. Bjork (2009) Arbitrage Theory in Continuous Time (3rd edition).

## Initial Remarks - Inst. Forwards

By definition the instantaneous forward is defined as

$$f(t,T) = -\frac{\partial \ln P(t,T)}{\partial T}$$

with its differential being

$$df(t,T) = -\frac{\partial}{\partial T} d \ln P(t,T) \tag{1}$$

and

$$P(t,T) = e^{-\int_t^T f(t,u)du}$$

## Initial Remarks - Models

We can broadly divide interest rate models in two categories

- **Endogenous Models:** or equilibrium models, make assumptions about economic variables and derive a process for the short rate, the market term structure is an output of the model (Vasicek, CIR etc.)
- **Exogenous Models:** or no arbitrage models, market term structure of interest rates is an input.

## Initial Remarks - Change of Measure

By the **Girsanov** theorem, if we have a Brownian motion $W_t$ under $\mathbb{Q}$, and we introduce a new process $y(t) = \int_0^t y_u du$ then $W^T = W_t - \int_0^t y_u du$ is a Brownian motion under $\mathbb{Q}^T$ defined via the Radon Nikodym derivative

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = e^{-\frac{1}{2}\int_0^t y_u^2 du + \int_0^t y_u dW_u} \tag{2}$$

in differential form

$$dW_t^T = dW_t - y_t dt$$

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## Framework

Heath, Jarrow and Morton (1992) assumed that, for a fixed a maturity $T$, the instantaneous forward rate $f(t, T)$ evolves, under a given measure, according to the following diffusion process:

$$df(t, T) = \mu(t, T)dt + \sigma_f(t, T)dW_t$$

## Framework

we will see that in HJM volatility dictates what kind of stochastic process the instantaneous forward takes

- the framework directly models the dynamics of instantaneous forward rates
- the framework provides an explicit relation between the volatility of the instantaneous forward rates and arbitrage-free drift
- the framework can be used to derive a limited [1] number of models, like extended short rate models and Libor Market Models (LMM)

[1] we will see later why models are "limited"

## Risk Neutral Dynamics

Under the HJM framework the short rate $r(t)$ is defined as the limit of the instantaneous forward $r(t) = f(t, t)$. We will start by modelling $P(t, T)$ (same maturity of $f(t, T)$), as the instantaneous forward $f(t, T)$ is not a traded asset

$$dP(t, T) = r_t P(t, T)dt + \sigma_P(t, T)P(t, T)dW_t \tag{3}$$

Starting from this process now we have to infer the dynamics of $f(t, T)$.

### Recall

We defined the instantaneous forward, its differential and ZCB price $P(t, T)$ in [1].

## Risk Neutral Dynamics

compute $d \ln P(t, T)$ by Ito

$$d \ln P(t, T) = \frac{1}{P(t, T)} dP(t, T) - \frac{1}{2} \frac{1}{P(t, T)^2} dP(t, T)^2 \tag{4}$$

in order to compute $\frac{1}{P(t,T)^2} dP(t, T)^2$, start from [3]:

$$\frac{dP(t, T)}{P(t, T)} = r_t dt + \sigma_P(t, T) dW_t \tag{5}$$

and add the quadratic term

$$\frac{dP(t, T)^2}{P(t, T)^2} = \sigma_P(t, T)^2 dt \tag{6}$$

## Risk Neutral Dynamics

Now substitute eq. [5], [6] in [4] and obtain

$$dlnP(t,T) = r_t dt + \sigma_P(t,T) dW_t - \frac{1}{2}\sigma_P(t,T)^2 dt$$

take derivative with respect to $T$ as we are interested in the instantaneous forward rate dynamics

$$\frac{\partial}{\partial T} dlnP(t,T) = \frac{\partial}{\partial T}\sigma_P(t,T) dW_t - \frac{1}{2} 2\sigma_P(t,T)\frac{\partial}{\partial T}\sigma_P(t,T) dt$$

## Risk Neutral Dynamics

By definition of the instantaneous forward rate [1] we can derive its dynamics under the risk neutral measure

$$df(t,T) = \sigma_P(t,T)\frac{\partial}{\partial T}\sigma_P(t,T)dt - \frac{\partial}{\partial T}\sigma_P(t,T)dW_t$$

with the **diffusion** coefficient of the instantaneous forward being

$$\sigma_f(t,T) = -\frac{\partial}{\partial T}\sigma_P(t,T)$$

$$\int_t^T \sigma_f(t,u)du + C = -\sigma_P(t,T)$$

## Risk Neutral Dynamics

we know that at maturity $\sigma(T, T) = 0$, therefore $C = 0$, we can write

$$\sigma_P(t, T) = \int_t^T \sigma_f(t, u) du$$

substitute to obtain the **drift** coefficient of the forward

$$\mu(t, T) = \sigma_f(t, T) \int_t^T \sigma_f(t, u) du$$

## Risk Neutral Dynamics

Finally, we get the full dynamics of $f(t, T)$ under the risk neutral measure $\mathbb{Q}$

$$df(t, T) = \left(\sigma_f(t, T) \int_t^T \sigma_f(t, u) du\right) dt + \sigma_f(t, T) dW_t \qquad (7)$$

## Forward Dynamics

The goal is to define another Brownian under the T-Forward probability measure and establish a connection with the previous.

### Change of Numeraire Toolkit!

Recall the change of numeraire trick [2]

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \frac{P(t,T)}{P(0,T)}\frac{B(0)}{B(t)} \tag{8}$$

[2] D. Brigo, F. Mercurio (2006) Interest Rate Models - Theory and Practice (2nd edition). Chapter 2

## Forward Dynamics

We already know the dynamics of $B(t)$

$$\frac{B(0)}{B(t)} = e^{-\int_0^t r_u du}$$

apply Ito to $lnP(t,T)$ to obtain the dynamics

$$\frac{P(t,T)}{P(0,T)} = e^{\int_0^t (r_u - \frac{1}{2}\sigma_P^2(u,T))du + \int_0^t \sigma_P(u,T)dW_u} \tag{9}$$

## Forward Dynamics

### Seems like the Radon-Nykodim derivative

If we compare the RN derivative [2] with the equation [9] we see that we established a connection as $y_u = \sigma_P^2(u, T)$.
We can write the new Brownian as

$$dW_t^T = dW_t - \sigma_P^2(t, T)dt = dW_t + \int_t^T \sigma_f(t, u)dudt$$

## Forward Dynamics

Now we can replace the new Brownian process in the Risk Neutral HJM [7] and obtain the dynamics under the new T-Forward probability measure

$$df(t,T) = \left(\sigma_f(t,T)\int_t^T \sigma_f(t,u)du\right)dt + \sigma_f(t,T)\left(dW_t^T - \int_t^T \sigma_f(t,u)dudt\right)$$

the drift term can be simplified as $f(t,T)$ is a martingale under the $\mathbb{Q}^T$ forward measure

$$df(t,T) = \sigma_f(t,T)dW_t^T \tag{10}$$

## Forward Dynamics

But how does this compare to the multi-curve framework?

- we have shown that it is possible to model a single T-maturity forward as a martingale using the zero coupon bond $P(t, T)$ numeraire
- in reality, you would have to model multiple forwards with maturity $T_1, T_2, \ldots, T_f$.

You don't want to model each one of them under a different probability measure. Typically, you can use the **longest maturity** $P(t, T_f)$ where $T_f > T$ where $T_f$ is the numeraire of the **Terminal forward measure**.

## Terminal Forward Dynamics

From the Girsanov theorem we get

$$dW_t^{T_f} = dW_t - \sigma_P^2(t, T_f)dt = dW_t + \int_t^{T_f} \sigma_f(t, u)dudt$$

if we substitute in the Risk Neutral equation [7] as we did before we get

$$df(t, T) = -\left(\sigma_f(t, T)\int_T^{T_f} \sigma_f(t, u)du\right)dt + \sigma_f(t, T)dW_t^T \quad (11)$$

## Different Measures

we can see now the difference between probability measures:

- $\mathbb{Q} \rightarrow$ the drift integral will be defined with the remaining maturity,
- $\mathbb{Q}^T \rightarrow$ drift will be zero,
- $\mathbb{Q}^{T_f} \rightarrow$ the drift integral will be defined between $T$ (maturity of the modelled forward) and $T_f$ (maturity of the terminal numeraire).

## Same Driver

### Remark

It doesn't matter which is the probability measure, **volatility drives the dynamics of** $f(t, T)$.

## Final Remarks

Finally, we need to do two final assumptions that simplify the model

- Deterministic $\sigma \rightarrow$ Then we know by the properties of Ito integral that $f \sim N$. We can see $dW$ term is Normally distributed, the dt term is deterministic, so the equation is a linear transformation of a combination of Normals. The result must be Normal.

## Final Remarks

- Separable volatility $\rightarrow$ the $dt$ term is deterministic so it's not going to dictate whether the process will be Markov. Everything depends on the the stochastic term. If the process is separable (it can be written as a product of functions of time and maturity)

$$\sigma_f(s,t) = g(s)h(T)$$

The short-rate process is not a Markov process in general. Notice in fact that the time $t$ appears in the stochastic integral both as extreme of integration and inside the integrand function. However, there are suitable specifications of $\sigma$ for which $r$ is indeed a Markov process. As proven by Carverhill (1994)

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## Short Rate Models

The theory of interest-rate modeling was originally based on the assumption of specific one-dimensional dynamics for the instantaneous spot rate process $r$.
Indeed, the existence of a risk-neutral measure implies that the arbitrage-free price at time t of a contingent claim with payoff $V(T)$ at time $T$ is given by

$$V(t) = \mathbb{E}_t[D(t,T)V(T)] = \mathbb{E}_t[e^{-\int_t^T r(s)ds}V(T)]$$

with the zero-coupon-bond price at time $t$ for the maturity $T$

$$P(t,T) = \mathbb{E}_t[e^{-\int_t^T r(s)ds}]$$

## Short Rate Models

From this last expression it is clear that whenever we can characterize the distribution of $e^{-\int_t^T r(s)ds}$ in terms of a chosen dynamics for $r$, conditional on the information available at time $t$, we are able to compute bond prices $P$.

## Short Rate Models

Let us assume that the instantaneous spot rate evolves under the real-world measure according to $\mathbb{Q}_0$

$$dr(t) = \mu(t, r(t))dt + \sigma(t, r(t))dW^{\mathbb{Q}_0}(t)$$

where $\mu$ and $\sigma$ are well behaved functions and $W^{\mathbb{Q}_0}(t)$ is a $\mathbb{Q}_0$ Brownian motion. It is possible to show[3] the existence of a stochastic process $\lambda$ such that if

$$dP(t, T) = \mu^T(t, r(t))dt + \sigma^T(t, r(t))dW^{\mathbb{Q}_0}(t)$$

then

$$\frac{\mu^T(t, r(t)) - r(t)P(t, T)}{\sigma^T(t, r(t))} = \lambda(t)$$

[3] see T. Bjork (2009) Arbitrage Theory in Continuous Time (3rd edition).

## Short Rate Models

for each maturity $T$, with $\lambda$ that may depend on $r$ but not on $T$. Moreover, there exists a measure $\mathbb{Q}$ that is equivalent to $\mathbb{Q}_0$ and is defined by the Radon-Nikodym derivative

$$\frac{d\mathbb{Q}}{d\mathbb{Q}_0} = e^{-\frac{1}{2}\int_0^t \lambda^2(u)du - \int_0^t \lambda(u)dW^{\mathbb{Q}_0}(u)} \tag{12}$$

As a consequence, the process $r$ evolves under $\mathbb{Q}$ according to

$$dr(t) = [\mu(t, r(t)) - \lambda(t)\sigma(t, r(t))]dt + \sigma(t, r(t))dW(t) \tag{13}$$

where $W(t) = W^{\mathbb{Q}_0}(t) + \int_0^t \lambda(u)du$ is a Brownian motion under $\mathbb{Q}$.

## Why Extended Models

A classic problem with the above models is their endogenous nature.
If we have the initial zero-coupon bond curve $T \to P^M(0, T)$ from the market, and we wish our model to incorporate this curve, we need forcing the model parameters to produce a model curve as close as possible to the market curve. Although the values $P^M(0, T)$ are actually observed only at a finite number of maturities, three parameters are not enough to reproduce satisfactorily a given term structure.

## Why Extended Models

These models cannot

- reproduce satisfactorily the initial yield curve,
- reproduce realistic volatility structures

, and so

## Derivation of Extended Models

To improve this situation, exogenous term structure models are usually considered. Such models are built by suitably modifying the above endogenous models. The basic strategy that is used to transform an endogenous model into an exogenous model is the inclusion of "time-varying" parameters.

## Derivation of Extended Models

We will focus on the derivation of the extended Vasicek model (Hull-White). The model we analyze implies a normal distribution for the short-rate process at each time.

## Derivation of Extended Models

A computation that will be helpful in the following slides is the instantaneous absolute volatility of instantaneous forward rates in affine models. In general

$$f(t,T) = -\frac{\partial}{\partial T} lnP(t,T),$$
$$df(t,T) = -\frac{\partial}{\partial T} lnA(t,T) + \frac{\partial}{\partial T} B(t,T) r(t),$$

so

$$df(t,T) = (\dots)dt + \frac{\partial}{\partial T} B(t,T)\sigma(t,r(t))dW(t),$$

where $\sigma(t,r(t))$ is the diffusion coefficient in the short rate dynamics.

## Derivation of Extended Models

It follows that the absolute volatility of the instantaneous forward rate $f(t, T)$ at time $t$ in a short rate model with an affine term structure is

$$\sigma_f(t, T) = \frac{\partial B(t, T)}{\partial T} \sigma(t, r(t)) \tag{14}$$

where $B(t, T)$ and $\sigma(t, r(t))$ are respectively the B function of the chosen short rate model and the volatility term.

## Derivation of Extended Models

Starting from eq.[14], the $B$ function in Vasicek is given by

$$B(t,T) = \frac{1}{\kappa}\left[1 - e^{-\kappa(T-t)}\right]$$

then

$$\frac{\partial}{\partial T} B(t,T) = e^{-\kappa(T-t)}$$

and $\sigma(t, r(t)) = \sigma$, therefore

$$\sigma_f(t,T) = \sigma e^{-\kappa(T-t)}$$

## Derivation of Extended Models

We know the dynamics of the HJM under the risk neutral measure $\mathbb{Q}$ being

$$df(t,T) = \left(\sigma_f(t,T)\int_t^T \sigma_f(t,u)du\right)dt + \sigma_f(t,T)dW_t$$

Substitute $\sigma_f(t,T)$ in the equation to get

$$df(t,T) = \sigma^2 e^{-\kappa(T-t)}\left(\int_t^T e^{-\kappa(u-t)}du\right)dt + \sigma e^{-\kappa(T-t)}dW_t$$

## Derivation of Extended Models

Do the necessary calculations

$$df(t,T) = \sigma^2 e^{-\kappa(T-t)} \left( -\frac{1}{\kappa} e^{-\kappa(u-t)} \Big|_{u=t}^{u=T} \right) dt + \sigma e^{-\kappa(T-t)} dW_t$$

$$df(t,T) = \sigma^2 e^{-\kappa(T-t)} \left( -\frac{1}{\kappa} \left( e^{-\kappa(T-t)} - 1 \right) \right) dt + \sigma e^{-\kappa(T-t)} dW_t$$

$$df(t,T) = \sigma^2 e^{-\kappa(T-t)} \left( \frac{1}{\kappa} \left( 1 - e^{-\kappa(T-t)} \right) \right) dt + \sigma e^{-\kappa(T-t)} dW_t$$

$$df(t,T) = \frac{\sigma^2}{\kappa} \left( e^{-\kappa(T-t)} - e^{-2\kappa(T-t)} \right) dt + \sigma e^{-\kappa(T-t)} dW_t$$

## Derivation of Extended Models

and integrate to obtain

$$\begin{aligned}\int_0^t df(s,T) =& \frac{\sigma^2}{\kappa}\int_0^t \left(e^{-\kappa(T-s)} - e^{-2\kappa(T-s)}\right) ds \\ &+ \sigma \int_0^t e^{-\kappa(T-s)} dW_s \\ f(t,T) - f(0,T) =& \frac{\sigma^2}{\kappa}\left(\frac{e^{-\kappa(T-s)}}{\kappa} - \frac{e^{-2\kappa(T-s)}}{2\kappa}\right)\Bigg|_{s=0}^{s=t} \\ &+ \sigma \int_0^t e^{-\kappa(T-s)} dW_s\end{aligned}$$

## Derivation of Extended Models

$$f(t,T) - f(0,T) = \frac{\sigma^2}{\kappa}\left(\frac{e^{-\kappa(T-t)} - e^{-\kappa T}}{\kappa} - \frac{e^{-2\kappa(T-t)} - e^{-2\kappa T}}{2\kappa}\right) + \sigma\int_0^t e^{-\kappa(T-s)}dW_s$$

as we know $r(t) = f(t,t)$, we can substitute to get

$$f(t,t) = f(0,t) + \frac{\sigma^2}{\kappa}\left(\frac{1-e^{-\kappa t}}{\kappa} - \frac{1-e^{-2\kappa t}}{2\kappa}\right) + \sigma\int_0^t e^{-\kappa(t-s)}dW_s$$

$$r(t) = f(0,t) + \frac{\sigma^2}{2\kappa^2}\left(1 - 2e^{-\kappa t} + e^{-2\kappa t}\right) + \sigma\int_0^t e^{-\kappa(t-s)}dW_s$$

## Derivation of Extended Models

we know that

$$dr(t) = df(t,T)\Big|_{T=t} + \frac{\partial}{\partial T} f(t,T)\Big|_{T=t} dt$$

where

$$\begin{aligned} df(t,T)\Big|_{T=t} &= \sigma dW_t \\ \frac{\partial}{\partial T} f(t,T)\Big|_{T=t} &= \frac{\partial}{\partial t} f(0,t) + \frac{\sigma^2}{\kappa}\left(e^{-\kappa t} - e^{-2\kappa t}\right) \\ &\quad - \kappa\sigma \int_0^t e^{-\kappa(t-s)} dW_s \end{aligned}$$

## Derivation of Extended Models

and get

$$dr(t) = \kappa \left( \frac{1}{\kappa} \frac{\partial}{\partial t} f(0,t) + \frac{\sigma^2}{\kappa^2} \left( e^{-\kappa t} - e^{-2\kappa t} \right) \right.$$
$$\left. - \sigma \int_0^t e^{-\kappa(t-s)} dW_s dt + \sigma dW_t \right.$$

where we factor out the $\kappa$ as we notice that the term $\sigma \int_0^t e^{-\kappa(t-s)} dW_s$ is the same as in equation [41].

## Derivation of Extended Models

if we isolate the term and we substitute we get

$$dr(t) = \kappa \left( \frac{1}{\kappa} \frac{\partial}{\partial t} f(0,t) + \frac{\sigma^2}{2\kappa^2} \left(1 - e^{-2\kappa t}\right) - r(t) + f(0,t) \right) dt + \sigma dW_t$$

## Derivation of Extended Models

Since we derived this settings from HJM we can conclude that this model will match the initial term structure, setting

$$\theta(t) = f(0,t) + \frac{1}{\kappa}\frac{\partial}{\partial t}f(0,t) + \frac{\sigma^2}{2\kappa^2}\left(1 - e^{-2\kappa t}\right)$$

we have the dynamics for the extended short rate model

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t$$

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## One Factor Gaussian Short Rate (HW)

We have shown that $r(t)$ evolves under the risk neutral measure according to

$$dr(t) = [\theta(t) - \kappa r(t)]dt + \sigma dW(t),$$

now consider $\kappa$ and $\sigma$ to be positive constants, with $\theta$ chosen to fit the currently observed yield curve in the market

$$\theta(t) = \frac{\partial}{\partial t} f^M(0,t) + \kappa f^M(0,t) + \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}),$$

where $\frac{\partial}{\partial t} f^M(0,t)$ denotes partial derivative of $f^M$ with respect to its second argument and $f^M(0,t)$ being the market instantaneous forward rate at time 0 for maturity $t$

## One Factor Gaussian Short Rate (HW) -Conditional Mean and Variance

The equation can be integrated to obtain

$$r(t) = r(s)e^{-\kappa(t-s)} + \alpha(t) - \alpha(s)e^{-\kappa(t-s)} + \sigma \int_s^t e^{-\kappa(t-u)} dW(u),$$

where

$$\alpha(t) = f^M(0,t) + \frac{\sigma^2}{2\kappa^2}(1 - e^{-\kappa t})^2,$$

$r(t)$ conditional on $r(s)$ with $t > s$ is Normally distributed with mean and variance given respectively by

$$\mathbb{E}[r(t)|r(s)] = r(s)e^{-\kappa(t-s)} + \alpha(t) - \alpha(s)e^{-\kappa(t-s)},$$

$$Var[r(t)|r(s)] = \frac{\sigma^2}{2\kappa}[1 - e^{-2\kappa(t-s)}].$$

## One Factor Gaussian Short Rate (HW) - Bonds

It's possible to obtain the price for zero-coupon bonds, allowing us to compute a new term structure as a function of a given short rate. It's shown that

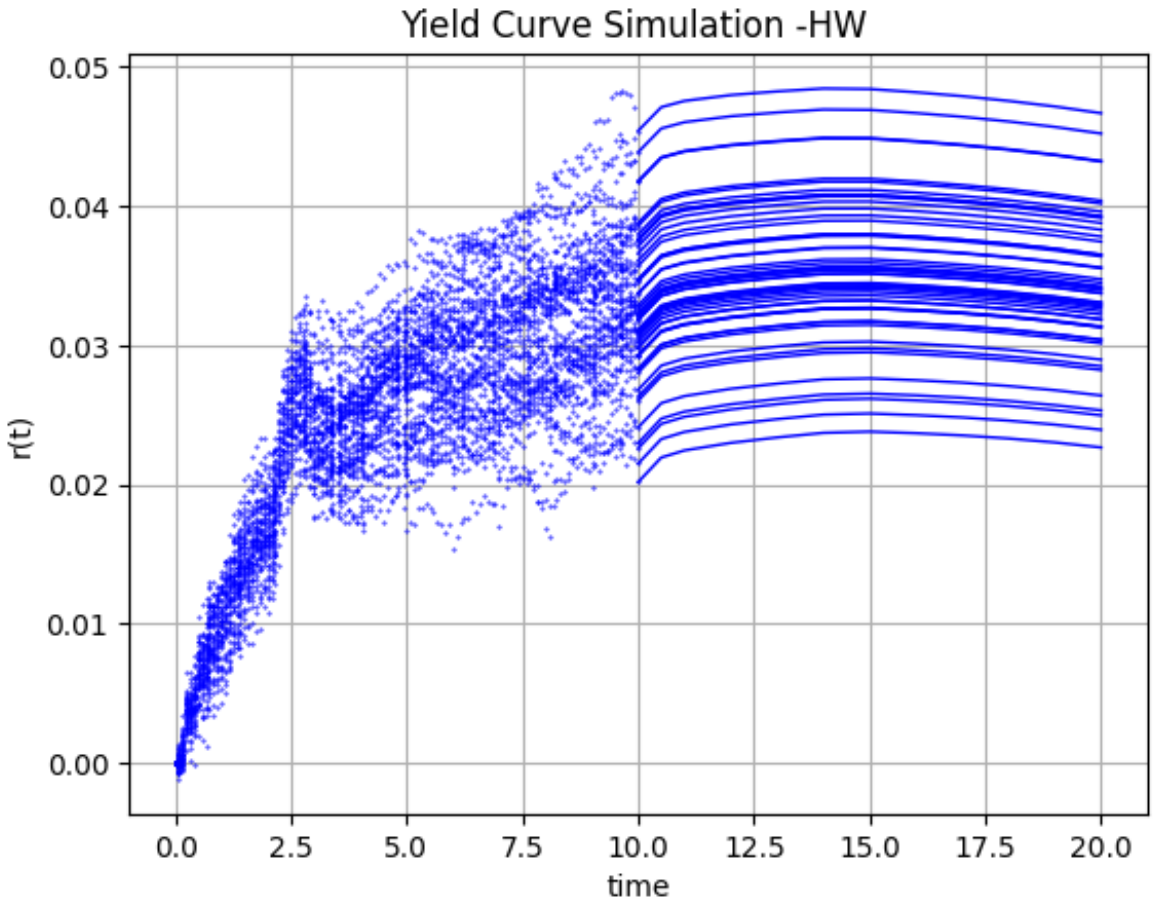
$$P(t,T) = A(t,T)e^{-B(t,T)r(t)}, \tag{15}$$

where

$$B(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa},$$

$$A(t,T) = \frac{P(0,T)}{P(0,t)} exp\left(B(t,T)f^M(0,t) - \frac{\sigma^2}{4\kappa}(1-e^{-2\kappa t})B(t,T)^2\right).$$

# Yield Curve Dynamics



## Two Factors Gaussian Short Rate (G2++)

It's not easy to find a realistic market situation that can be correctly represented by a one factor model, in such models it's impossible to find any feasible opportunities to capture interest rates correlation.
It is shown[4] that a model with two or more factors, capable to include forward rates correlation, produces better results and is ideal to explain an acceptable portion of interest rates variability.

[4] Rebonato

## Two Factors Gaussian Short Rate (G2++)

The Gaussian two factor model defines the short rate as the sum of two correlated Gaussian random variables, under the risk neutral measure $\mathbb{Q}$ the interest rate dynamics are

$$r(t) = x(t) + y(t) + \varphi(t), \qquad r(0) = r_0.$$

With $\varphi(t)$ being a deterministic function of time, chosen to exactly fit the initial term structure of interest rates, $\varphi(0) = r_0$

$$\begin{aligned}\varphi(t) =& f(0,t) + \frac{\sigma^2}{2a^2}(1 - e^{-at})^2 \\ &+ \frac{\eta^2}{2b^2}(1 - e^{-bt2}) + \rho\frac{\sigma\eta}{ab}(1 - e^{-at})(1 - e^{-bt}),\end{aligned}$$

## Two Factors Gaussian Short Rate (G2++)

and the two processes $x(t)$ and $y(t)$ defined as

$$\begin{aligned} dx(t) &= -ax(t)dt + \sigma dW_1(t), \qquad x(0) = 0; \\ dy(t) &= -by(t)dt + \eta dW_2(t), \qquad y(0) = 0; \end{aligned}$$

where $W_1$ and $W_2$ are two different Brownian motion with correlation $\rho$

$$dW_1(t)dW_2(t) = \rho dt.$$

## Two Factors Gaussian Short Rate (G2++)

If we integrate we obtain the two stochastic differential equations solutions

$$\begin{aligned} x(t) =& x(s)e^{-a(t-s)} + \sigma \int_s^t e^{-a(t-u)} dW_1(u), \\ y(t) =& y(s)e^{-b(t-s)} + \eta \int_s^t e^{-b(t-u)} dW_2(u); \end{aligned}$$

then the short rate evaluated at $t_0$ for a given maturity $T$ is expressed as

$$r_0(T) = \sigma \int_0^T e^{-a(T-u)} dW_1(u) + \eta \int_0^T e^{-b(T-u)} dW_2(u) + \varphi(T);$$

letting $T \to \infty$ the two factors reverts to their initial value $x(\infty), y(\infty) \to 0$ and the $r(T)$ process mean reverts to $\varphi(t)$.

## Two Factors Gaussian Short Rate (G2++)

By the properties of stochastic integral of deterministic functions we obtain mean, variance and covariance from the bivariate Normal distribution of $r(t)$ conditional on $r(s)$ with $s < t$ as

$$\mathbb{E}^{\mathbb{Q}}[x(t)|x(s)] = x(s)e^{-a(t-s)}, \quad \mathbb{E}^{\mathbb{Q}}[y(t)|y(s)] = y(s)e^{-b(t-s)};$$

$$Var^{\mathbb{Q}}[x(t)] = \frac{\sigma^2}{2a}[1 - e^{-2a(t-s)}], \quad Var_s^{\mathbb{Q}}[y(t)] = \frac{\eta^2}{2b}[1 - e^{-2b(t-s)}];$$

$$cov_s[x(t), y(t)] = \rho\frac{\sigma\eta}{a+b}[1 - e^{-(a+b)(t-s)}].$$

## Two Factors Gaussian Short Rate (G2++)

The two processes can also be rewritten in terms of two independent Brownian motions $\tilde{W}_1$ and $\tilde{W}_2$ applying a Cholesky decomposition

$$dW_1(t) = d\tilde{W}_1,$$
$$dW_2(t) = \rho d\tilde{W}_1 + \sqrt{1-\rho^2} d\tilde{W}_2.$$

Substituting in the two equation for $x(t)$ and $y(t)$ by integration we get

$$x(t) = x(s)e^{-a(t-s)} + \sigma \int_0^t e^{-a(t-u)} d\tilde{W}_1(u),$$
$$y(t) = y(s)e^{-b(t-s)} + \eta\rho \int_0^t e^{-b(t-u)} d\tilde{W}_1(u) + \eta\sqrt{1-\rho^2} \int_0^t e^{-b(t-u)} d\tilde{W}_2(u).$$

## Two Factors Gaussian Short Rate (G2++)

From the short rate we obtain the discount bond term structure

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_s ds}\right];$$

where the integral in the exponential $I(t,T) = \int_t^T r_s ds$ is shown[5] to be Normally distributed with mean given by

$$\mathbb{E}_t^{\mathbb{Q}}[I(t,T)] = \frac{1-e^{-a(T-t)}}{a}x(t) + \frac{1-e^{-b(T-t)}}{b}y(t),$$

[5] see D. Brigo, F. Mercurio (2006) Interest Rate Models - Theory and Practice (2nd edition)

## Two Factors Gaussian Short Rate (G2++)

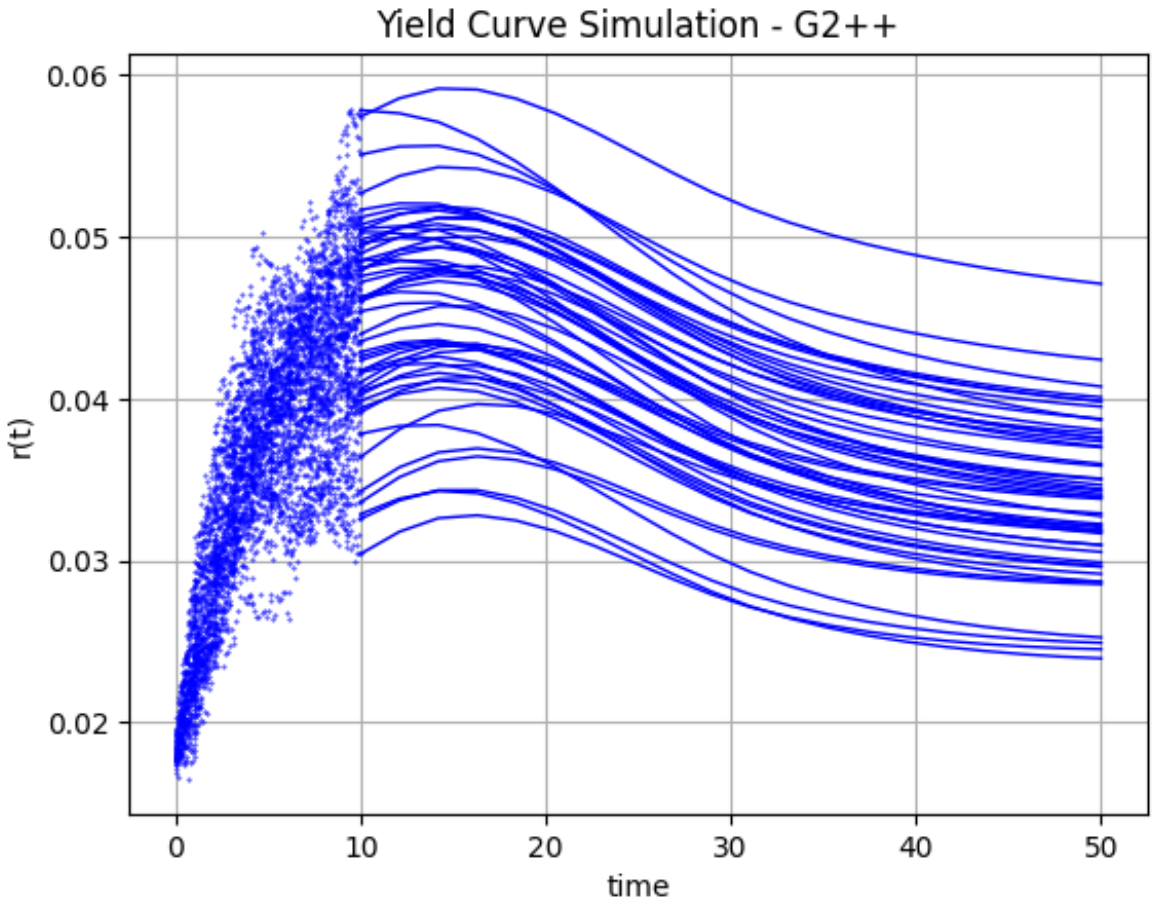
and variance

$$
\begin{aligned}
Var^{\mathbb{Q}}[I(t,T)] = & \frac{\sigma^2}{a^2}\left[T-t+\frac{2}{a}e^{-a(T-t)}-\frac{1}{2a}e^{-2a(T-t)}-\frac{3}{2a}\right] \\
& \frac{\eta^2}{b^2}\left[T-t+\frac{2}{b}e^{-b(T-t)}-\frac{1}{2b}e^{-2b(T-t)}-\frac{3}{2b}\right] \\
& 2\rho\frac{\sigma\eta}{ab}\left[T-t+\frac{e^{-a(T-t)}-1}{a}+\frac{e^{-b(T-t)}-1}{b}\right. \\
& \left.-\frac{e^{-(a+b)(T-t)}-1}{a+b}\right].
\end{aligned}
$$

## Two Factors Gaussian Short Rate (G2++)

With the zero coupon bond price at time t being defined as

$$P(t,T) = \frac{P(0,T)}{P(0,t)} e^{A(t,T)},$$

$$A(t,T) = \frac{1}{2}[var(t,T) - var(0,T) + var(0,t)] - \mathbb{E}_t^{\mathbb{Q}}[I(t,T)].$$

# Yield Curve Dynamics